

4) $u_{tt} = u_{xx} + e^{-t}$

* $u(x,t) = A_0(t) + \sum_{n=1}^{\infty} a_n(t) \cos(n\pi x)$ *

1. $A_0''(t) + \sum_{n=1}^{\infty} a_n''(t) \cos(n\pi x) = \sum_{n=1}^{\infty} a_n(t) (-n\pi)^2 \cos(n\pi x) + e^{-t}$

1 = $\sum_{n=1}^{\infty} c_n \sin(n\pi x)$

$\sum_{n=1}^{\infty} c_n A_0''(t) \sin(n\pi x) + \sum_{n=1}^{\infty} a_n''(t) \cos(n\pi x) = \sum_{n=1}^{\infty} a_n(t) (-n\pi)^2 \cos(n\pi x) + \sum_{n=1}^{\infty} c_n e^{-t} \sin(n\pi x)$

* $\Rightarrow \begin{cases} A_0''(t) = e^{-t} \\ a_n''(t) = -n^2 \pi^2 a_n(t) \end{cases}$ * $u_t(x,0) = 0$

Aside: $\sum_{n=0}^{\infty} c_n \cos(n\pi x) = 0$
 only because
 (i) $n\pi x$
 (ii) the interval of the problem



* $1 = \sum_{n=0}^{\infty} c_n \cos(n\pi x)$
 $= c_0 + \sum_{n=1}^{\infty} c_n \cos(n\pi x)$
 $1 = c_0$
 • $1 = \sum_{n=0}^{\infty} c_n \delta_{n0} \cos(n\pi x)$

$\sum_{n=0}^{\infty} c_n A_0''(t) \delta_{n0} \cos(n\pi x) + \sum_{n=1}^{\infty} c_n \cos(n\pi x) a_n''(t)$
 $= \sum_{n=1}^{\infty} a_n(t) (-n\pi)^2 c_n \cos(n\pi x) + \sum_{n=0}^{\infty} c_n \delta_{n0} e^{-t} \cos(n\pi x)$

$\sum_{n=0}^{\infty} c_n \delta_{n0} \cos(n\pi x) [A_0''(t) - e^{-t}] + \sum_{n=1}^{\infty} c_n \cos(n\pi x) [a_n''(t) - a_n(t) (-n\pi)^2] = 0$

$$L \rightarrow \begin{cases} A''(t) = e^{-t} & (a) \\ a''(t) = a_n(t) [-(n\pi)^2] = 0 & (b) \end{cases}$$

$$(a) A''(t) = e^{-t}$$

$$A'(t) = -e^{-t} + A$$

$$\star A(t) = e^{-t} + Dt + E \star$$

$$(b) a''(t) = -(n\pi)^2 a_n(t)$$

$$a(t) = \tilde{A} \cos(n\pi t) + \tilde{B} \sin(n\pi t)$$

$\Rightarrow$ I.C.:

$$u(x, 0) = X \leftarrow$$

$$u_t(x, 0) = 0 \star$$

$$u(x, t) = A_0(t) + \sum_{n=1}^{\infty} a_n(t) \cos(n\pi x)$$

$$u_t(x, t) = A_0'(t) + \sum_{n=1}^{\infty} a_n'(t) \cos(n\pi x)$$

$$u_t(x, 0) = 0$$

$$\hookrightarrow A_0'(0) = 0$$

$$a_n'(0) = 0$$

$$(a) A_0'(0) = 1 + D = 0$$

$$(b) a_n'(0) = -\tilde{A} \sin(0) n\pi + \tilde{B} n\pi \cos(0) = 0$$

$$\tilde{B} = 0$$

$$(a) A_0(t) = e^{-t} + t + E$$

$$(b) a_n(t) = \tilde{A} \cos(n\pi t)$$

Final I.C.: $u(x, 0) = X$

$$u(x, t) = e^{-t} - t + E + \sum_{n=1}^{\infty} (\tilde{A} \cos(n\pi t) \cos(n\pi x))$$

$$u(x, 0) = 1 + E + \sum_{n=1}^{\infty} \tilde{A} \cos(n\pi x) = X$$

$$u_t(x,0) = \underbrace{A_0'(0)}_{=0} + \sum_{n=1}^{\infty} \underbrace{a_n'(0)}_{=0} \cos(n\pi x)$$

$$\Rightarrow A_0'(0) = 0$$

$$a_n'(0) = 0$$

* Idea: we should apply our I.C.s to u before we plug back into PDE

Other I.C.: $u(x,0) = x$

$$\underbrace{A_0(t)}_{=1} + \sum_{n=1}^{\infty} \underbrace{a_n(t)}_{=0} \cos(n\pi x) = \underbrace{\sum_{n=0}^{\infty} a_n(t) \cos(n\pi x)}_{\text{easier to solve for } a_n(t)}$$

$A_0 = a_0$

I.C.: $u(x,0) = \sum_{n=0}^{\infty} a_n(0) \cos(n\pi x) = x$ *

* mult. by $\cos(n\pi x)$, integrate

$$\underline{a_n(0)} = 2 \int_0^1 x \cos(n\pi x) dx$$

$$n \neq 0 \quad \boxed{a_n(0) = \frac{2((-1)^n - 1)}{n^2 \pi^2}}$$

$$\underline{n=0}: a_0(0) = \int_0^1 x dx = \frac{1}{2}, \quad \boxed{A_0(0) = \frac{1}{2}}$$

* Now, plug into PDE:

$$A_0''(t) = e^{-t}$$

$$a_n''(t) = -(n\pi)^2 a_n(t)$$

$$A_0(0) = \frac{1}{2}$$

a) $A_0(t) = e^{-t} + t + E$

$A_0(t) = 1 + E = \frac{1}{2}$

$$A_0(0) = 1 + E = -\frac{1}{2}$$

$$\Rightarrow E = -1/2$$

$$A_0(t) = e^{-t} + t - \frac{1}{2} \quad *$$

$$(b) \quad a_n(t) = \hat{A} \cos(n\pi t)$$

$$a_n(0) = \hat{A} = 2 \frac{((-1)^n - 1)}{(n\pi)^2}$$

$$a_n(t) = \frac{2((-1)^n - 1)}{(n\pi)^2} \quad *$$

$$u(x, t) = A_0(t) + \sum_{n=1}^{\infty} a_n(t) \cos(n\pi x)$$

